BASIC THERMODYNAMICS

Basic Equations

$$dU = T\,dS - P\,dV + \mu\,dN \qquad U = TS - PV + \mu N$$

$$d\mu = -s\,dT + v\,dP \qquad d\left(\frac{\mu}{T}\right) = u\,d\left(\frac{1}{T}\right) + v\,d\left(\frac{P}{T}\right)$$

Ideal Gas

$$U = cNk_BT$$

$$PV = Nk_BT$$

$$S = Nk_B \log(vu^c) + Ns_0$$

van der Waals Fluid

$$u + \frac{a}{v} = ck_BT$$

$$P = \frac{k_BT}{v-b} - \frac{a}{v^2}$$

$$S = Nk_B \log\left((v-b)\left(u + \frac{a}{v}\right)^c\right) + Ns_0$$

Second Derivatives

$$\alpha \equiv \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P \qquad c_v \equiv \frac{T}{N}\left(\frac{\partial S}{\partial T}\right)_V \qquad \kappa_T \equiv -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$$

$$c_P \equiv \frac{T}{N}\left(\frac{\partial S}{\partial T}\right)_P \qquad \kappa_s \equiv -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_S$$

Kinetic Theory

$$P(v_i) = \left(\frac{m}{2\pi kT}\right)^{1/2} \exp\left(-\frac{mv_i^2}{2kT}\right)$$

$$P(v) = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2} v^2 \exp\left(-\frac{mv^2}{2kT}\right)$$

In this space, I use a very nicely organized chart of 18 Maxwell Relations organized by thermodynamic potential from *Thermodynamics and an Introduction to Thermostatistics*, 2nd ed., by Herbert B. Callen, John Wiley & Sons, 1985, Section 7-1, pg 182-183

REDUCTION OF DERIVATIVES

$$\left(\frac{\partial X}{\partial Y}\right)_Z = -\frac{\left(\frac{\partial Z}{\partial Y}\right)_X}{\left(\frac{\partial Z}{\partial X}\right)_Y}$$

$$c_v = c_P - \frac{TV\alpha^2}{N\kappa_T}$$

PHASE TRANSITIONS

Stability

$$\frac{\partial^2 S}{\partial U^2} \le 0 \qquad \frac{\partial^2 S}{\partial V^2} \le 0 \qquad \frac{\partial^2 S}{\partial U^2}\frac{\partial^2 S}{\partial V^2} - \left(\frac{\partial^2 S}{\partial U \partial V}\right)^2 \ge 0$$

$$\frac{\partial^2 U}{\partial S^2} \ge 0 \qquad \frac{\partial^2 U}{\partial V^2} \ge 0 \qquad \frac{\partial^2 U}{\partial S^2}\frac{\partial^2 U}{\partial V^2} - \left(\frac{\partial^2 U}{\partial S \partial V}\right)^2 \ge 0$$

Clausius-Clapeyron

$$\left(\frac{dP}{dT}\right)_{cc} = \frac{\Delta S}{\Delta V} = \frac{\ell}{T\Delta v}$$

GAUSSIAN INTEGRALS

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2} \qquad \int_0^\infty x\,e^{-x^2}\,dx = \frac{1}{2}$$

$$\int_0^\infty x^2\,e^{-x^2}\,dx = \frac{\sqrt{\pi}}{4} \qquad \int_0^\infty x^3\,e^{-x^2}\,dx = \frac{1}{2}$$

STATISTICAL MECHANICS

$$\ln N! = N\ln N - N$$

$$\langle f\rangle = \sum_x f(x)P(x) \qquad \langle f\rangle = \int_{-\infty}^{\infty} f(x)P(x)\,dx$$

$$\sigma_f = \sqrt{\langle f^2\rangle - \langle f\rangle^2}$$

Boltzmann Factor

$$e^{-\beta E}$$

Gibbs Factor

$$e^{-\beta(E-N\mu)}$$

Formalism	Partition Function	Link Equation
Microcanonical	$\Omega = \sum_{\text{states } j} 1$	$S(U,V,N) = k_B \ln\Omega$
Canonical	$Z = \sum_{\text{states } j} e^{-\beta E_j}$	$-\frac{F(T,V,N)}{T} = k_B \ln Z$
Grand Canonical	$\mathcal{Z} = \sum_{\text{states } j} e^{-\beta(E_j - N_j\mu)}$	$-\frac{\Psi(T,V,\mu)}{T} = k_B \ln \mathcal{Z}$

Other

$$U = -\frac{\partial}{\partial\beta}(\ln Z) = k_B T^2 \frac{\partial}{\partial T}(\ln Z)$$

$$N = k_B T \frac{\partial}{\partial\mu}(\ln \mathcal{Z})$$

$$U = k_B T^2 \frac{\partial}{\partial T}(\ln \mathcal{Z}) + k_B T\mu \frac{\partial}{\partial\mu}(\ln \mathcal{Z})$$

Planck Distribution

$$f(\omega) = \frac{1}{e^{\beta\hbar\omega} - 1}$$

Fermi/Bose Distributions

$$f_{\substack{\text{fermion}\\\text{boson}}}(\varepsilon) = \frac{1}{e^{\beta(\varepsilon-\mu)} \pm 1}$$